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WildfireIQ Kamloops: An Open Platform for Wildfire Risk, Air Quality, Climate Trends, and Community Preparedness in British Columbia, with a Data-Grounded Assistant

Gursahib Singh[1], Deeparsh Singh Dang[1], and Ghazanfar Latif[2,*]

[1] Department of Computing Science, Thompson Rivers University, Kamloops, BC, Canada; gursahib99888@gmail.com; deeparshsingh10@gmail.com

[2] Department of Computing Science, Thompson Rivers University, Kamloops, BC, Canada; glatif@tru.ca

* Correspondence: glatif@tru.ca

Abstract

Background and objectives. Wildfire seasons in British Columbia have grown in scale and consequence, and the Thompson-Okanagan region around Kamloops has been among the most affected in Canada over the past decade. The 2023 season was the largest on record nationally. Smoke exposure is now a recurring public-health event, yet the information available to residents remains scattered: air-quality apps do not connect to fire activity, evacuation notices are static, and no public tool ties fire risk, smoke, long-term climate change and household preparedness together for this region. WildfireIQ Kamloops set out to close that gap with one open, free-to-run web platform built entirely from public data, and to do so with transparent, held-out validation of every prediction it shows.

Methods. The platform is built as a single backend process (Python, FastAPI) and a browser front end (React, TypeScript, CesiumJS). Seventeen scheduled jobs pull from the BC Wildfire Service, BC Emergency Management and Climate Readiness, NASA FIRMS, Environment and Climate Change Canada, Open-Meteo and WAQI on cadences from five minutes to nightly, writing one Parquet file per dataset; the API serves from those files, so the platform keeps working when a source is down. Two models were trained. The wildfire-risk model is a single LightGBM classifier pooled across four regions (Thompson-Okanagan, Central Okanagan, Lower Mainland, Prince George), predicting the probability that at least one fire ignites in a region on a given day from that region's ERA5 weather, Canadian Fire Weather Index codes computed with the Van Wagner equations, lagged and rolling weather, drought signals and a base-rate prior; the regional probability is scaled by each H3 hexagon's historical fire density into a four-class map. It was trained on 1999–2021, calibrated on 2022, and tested on a fully unseen 2023, with results reported per region. The air-quality model is 21 LightGBM quantile regressors (seven horizons to 48 hours, three quantiles) over 469 days of hourly CAMS PM2.5 and co-located weather at Kamloops; its q10-q90 band is conformalised so that its stated coverage is a measured property rather than a nominal one. A fifth module was added after the four planned features were complete: an assistant answering only through 25 read-only tools over the platform's own data, primed with a live situation brief, using GLM 5.3 Flash via OpenRouter behind rate and spend limits and a deterministic emergency-wording tripwire.

Results. All five modules are built, connected to live data, and covered by 167 backend and 36 frontend tests. On the unseen 2023 season the risk model reaches a precision-recall AUC of 0.72 for the Thompson-Okanagan, 0.61 for Prince George and 0.51 for the Central Okanagan against a Fire-Weather-Index baseline of 0.37 and a climatology baseline of 0.19; the Lower Mainland, a low-event area with 35 fire-days in the test year, scores

0.29 and is disclosed as such. Pooled ROC-AUC is 0.874 and the calibrated Brier score 0.108. The air-quality forecaster, retrained on a 469-day record that now spans a full smoke season, beats a persistence baseline at every horizon from six hours out by 6 to 22 per cent, and loses below that, where hour-to-hour autocorrelation makes repeating the last reading hard to improve on. Its uncertainty band covered 59 to 68 per cent of held-out observations while implying 80; conformalised with a per-horizon widening factor, it now measures 79 to 81. Over 1999–2025 the Thompson-Okanagan record shows a July mean daily-maximum temperature trend of +0.18 °C per year (95% CI 0.06–0.32), rising vapour-pressure deficit, and +1.95 high-fire-danger days per year (95% CI 1.0–2.75), all distinguishable from noise by Theil-Sen estimation with bootstrap intervals; summer precipitation, area burned and season length are not. The assistant passes a 32-case live evaluation spanning every data surface and a set of misuse attempts, at a measured $0.0006 to $0.0013 and a median of six to fourteen seconds per answer across repeated sweeps.

Conclusions. WildfireIQ shows that one small team, using only free and open data, can deliver a validated, honest wildfire information platform for a region, and that a language-model assistant can be made useful and safe by grounding it exclusively in the platform's own tools rather than its training data. Regions are configuration rather than code: adding a city to the risk model is a data step, and the components tied to British Columbia are enumerated so the platform can be carried to other provinces. Known limits are stated in the software itself rather than left for a reader to discover: one weather series per region; a single-point air-quality forecast whose calibrated band rests on a record containing a single severe smoke season, so its widening factor is recomputed on every retrain and coverage will dip when a season opens outside that record; a placeholder in place of the live climate-model ensemble; and a platform that is informational and never a substitute for the BC Wildfire Service or emergency officials. The code, models, model cards and documentation are released under the MIT licence.

Keywords: wildfire risk prediction; air quality forecasting; Fire Weather Index; LightGBM; quantile regression; H3 hexagonal grid; community preparedness; FireSmart; climate trends; Theil-Sen; LLM agent; tool use; British Columbia; Thompson-Okanagan; environmental informatics; open data

The platform at a glance

Item	Figure
Modules	Globe · Air Quality · Preparedness · Climate Trends · Assistant
Public data sources	BC Wildfire Service, BC EMCR, NASA FIRMS, ECCC GeoMet, Open-Meteo, WAQI
Scheduled data jobs	17 (15 recurring, every 5 minutes to nightly; 2 one-time)
REST endpoints	34, each response carrying its source and capture time
Historical fire records	96,356 incidents province-wide, 1999 to today
Weather history	27 years of ERA5 reanalysis per region, extended daily
Risk model	Pooled LightGBM, 4 regions, 523 H3 hexagons, 42 features
Held-out 2023 PR-AUC by region	0.72 · 0.61 · 0.51 · 0.29 (FWI baseline 0.37)
Air-quality model	21 LightGBM quantile models to 48 h; conformalised band, 79-81% coverage
Assistant	25 tools, GLM 5.3 Flash via OpenRouter, 32/32 live evaluation
Automated tests	167 backend, 36 frontend

Source code, model cards and the full documentation set: github.com/DeeparshSingh/WildfireIQ. Every figure above is taken from the released build of 8 September 2026.